

SURGICAL PATHOLOGY REPORT

*** ADDENDUM REPORT ***

ADDENDUM REPORT: NUMBER TWO

Right colon, partial colectomy:

Poorly differentiated adenocarcinoma, deficient gene expression for MLH-1 and PMS-2; intact gene expression for MSH-2 and MSH-6 (immunohistochemistry).
See comment.

COMMENT: The tumor is DNA mismatch repair deficient, and should also exhibit high frequency microsatellite instability. Most tumors with loss of MLH1 and PMS2 protein expression have primary underlying defects in MLH1 expression, with co-existent loss of expression of the binding partner protein PMS2. In most cases (about 90%) MLH1 loss is caused by somatic hypermethylation of the promoter region of the MLH1 gene. This is a common mechanism involved in the progression of serrated neoplasms of the right colon. Some individuals with these tumors may have an inherited predisposition to the formation of multiple serrated neoplasms (hyperplastic/serrated polyposis), although the genetic basis of this syndrome is unknown. These individuals are at increased risk of metachronous cancers of the right colon.

About 10% of MLH1 deficient tumors are caused by inherited germline defects in the MLH1 gene (Lynch syndrome; also known as hereditary non-polyposis colorectal cancer or HNPCC). The possibility of an underlying PMS2 germline defect cannot be excluded. The tumor does not reveal a BRAF mutation (see below) which is often seen with somatic or hypermethylation of the promoter region of the MLH1 gene. But to further clarify a block of tumor is submitted to the Mayo Clinic Laboratory for hypermethylation testing.

Family history evaluation and additional testing could be helpful in determining if MLH1 loss in the current tumor is caused by methylation or germline mutation. Genetic counseling and testing should be considered if clinically indicated.

DNA mismatch repair deficient cancers are associated with improved overall survival. Some studies also demonstrate resistance to 5-FU based chemotherapy and sensitivity to irinotecan (CPT-11).

ADDENDUM REPORT: NUMBER ONE

BRAF Mutation Analysis

SPECIMEN SOURCE: COLON

PARAFFIN BLOCK NUMBER: IS11-22551, 6

BRAF Mutation Analysis: NOT DETECTED

This result was reviewed and interpreted by 

Based on sequence analysis, no mutations were detected in BRAF exon 11, 12 and 15.

Cancer-activating BRAF mutations can be found in 14% of colorectal cancer, 2% of lung cancer, and 40% of thyroid cancer. Presence of BRAF mutation in colorectal cancer has been reported to render the tumor resistant to anti-EGFR therapy.

Documented BRAF polymorphism is usually not reported and is a neutral variant (no amino acid change).

This assay may not be able to detect mutation if the tumor constitutes less than 15-20% of analyzed sample. However, we specifically microdissect tumor and selectively analyze the tumor in the provided tissue.

The BRAF mutation analysis from Quest Diagnostics uses a DNA-based PCR-sequencing assay to detect somatic mutations on exon 11, 12, and 15 of BRAF, which cover 102 out of 104 reported mutations (99.6% incidence).

This test was developed and its performance characteristics 
Performance characteristics refer to the analytical performance of the test.

BRAF Mutation Analysis

SPECIMEN SOURCE: COLON

PARAFFIN BLOCK NUMBER:

No Prior-treatment



BRAF Mutation Analysis: TNP
TNP-TEST REQUEST CANCELLED - NO CHARGE.

NORMAL TISSUE: BLOCK FOR MSI

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This test was developed and its performance characteristics have been determined by [REDACTED]
Performance characteristics refer to the analytical performance of the test. Test performed at [REDACTED]

DIAGNOSIS

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Right colon, partial colectomy:

Tumor Characteristics:

1. Histologic type: Adenocarcinoma.
2. Histologic grade: Poorly differentiated.
3. Tumor site: Cecum.
4. Tumor size: 12.9 x 5.0 x 1.5 cm.
5. Tumor subtype: Mucinous tumor component present.
6. Tumor extension: Tumor invades through muscularis propria into adjacent fat to involved inked surface.
7. Margins: Proximal margin (small bowel) - negative; distal margin - negative; circumferential - (radial) margin - positive (tumor extends to inked surface).
8. Lymphovascular invasion: Present.

Lymph Node Status:

1. Nine of sixteen lymph nodes demonstrating metastatic, poorly differentiated carcinoma (9/16).
2. Largest replaced lymph node measures up to 1.4 cm.

Other:

1. pTNM stage: pT4a, pN2b.

COMMENTS:

Multiple sections of the very large tumor reveals a poorly differentiated adenocarcinoma, composed predominantly of solid nests of cells having a moderate amount of cytoplasm and large vesicular nuclei with prominent nucleoli. Some rudimentary gland formation is evident. Elsewhere, there is a signet ring appearance to the cells and also lakes of mucin can be identified. The tumor invades transmurally into the adjacent fat and focally to the inked surface of the fat (presumed radial margin). Extensive angiolymphatic invasion is seen within the tumor. Multiple lymph nodes are involved by metastatic carcinoma (see above).

Due to the poorly differentiated appearance of the neoplasm, a series of immunoperoxidase stains are performed. The tumor shows moderate to focally strong immunoreactivity for CK7. A stain for CK20 is negative. The CDX immunostain reveals some focal immunoreactivity within tumor cell nuclei. Stains for CK5-6, synaptophysin, PSA, chromogranin and TTF-1 are negative. A stain for calretinin is negative also (no nuclear reactivity). This pattern of immunostaining is somewhat unusual for a colorectal primary, since most show CK20 immunoreactivity and are negative for CK7. However, the focal CDX2 positivity is supportive of a colorectal primary. In addition, the poorly differentiated appearance of neoplasm and the right sided location suggests a possible MSI (microsatellite instability) high type of lesion. MSI high carcinomas often do show a reverse staining pattern for CK7 and CK20.

Immunohistochemical stains for mismatch repair enzymes (evaluation for HNPCC) and/or MSI-PCR testing will be performed on the tumor and reported as an addendum.

The groin lesion was reviewed as is a squamous cell carcinoma of the skin.

CLINICAL INFORMATION

CLINICAL HISTORY:

Preoperative Diagnosis: Colon carcinoma.

Postoperative Diagnosis:

Symptoms/Radiologic Findings:

SPECIMENS:

Right colon

SPECIMEN DATA

GROSS DESCRIPTION:

The specimen is received in a single formalin filled container labeled with the patient's name [REDACTED] and consists of a right hemicolectomy specimen comprised of terminal ileum (11.5 cm in length x 1.8 cm in diameter), cecum/right colon (21.0 cm in length and ranging from 4.0 up to 7.0 cm in diameter), and vermiform appendix (4.3 cm in length and ranging from 0.5 to 0.7 cm in diameter). The specimen is received stapled closed at both ends. The serosa is tan-pink and glistening with adherent yellow-tan mesocolonic fat and a portion of omentum. The cecal pouch is remarkable for a 12.8 x 5.0 x 1.5 cm circumferential fungating mass that approaches to within 11.0 cm of the ilial margin and 15.5 cm of the colonic margin. On section, this mass extends through the muscularis and approaches to within 0.1 cm of the inked deep margin. The remainder of the mucosa is pink-tan and glistening with the normal folds. No additional lesions are identified. Sections through the appendix reveal a 0.3 cm outpouching at the distal tip. The lumen is pinpoint with no perforations or fecaliths.

Sections through the omentum reveal a yellow-tan fibrofatty cut surface. No tumor is identified. Within the mesocolonic fat are multiple firm fatty possible lymph nodes ranging from 0.5 up to 2.2 cm in greatest dimension. Many are involved by gray-white tumor. Representative sections are submitted in cassettes 1 through 19 labeled [REDACTED] designated as follows: 1) colonic margin, en face; 2) ileal margin, en face; 3-6) mass to inked deep margin; 7) mass to normal; 8) vermiform appendix; 9 and 10) five whole possible lymph nodes in each cassette; 11) four whole possible lymph nodes; 12) two whole possible bisected lymph nodes (one inked); 13-15) one whole possible bisected lymph node in each cassette; 16-19) one whole possible quadrisectioned lymph node.

Additionally, a yellow, green, and blue cassette are submitted for genetic [REDACTED]

Additional representative sections are submitted as labeled: 20-23 mass to uninvolved mucosa. The blocks are labeled [REDACTED]